PAPER_711: NGC 2014 + NGC 2020 Variant 2: Wolf-Rayet Cone UQFF Analysis

Class: `NGC2014NGC2020Variant2UQFF` **CP4 Entry:** #295 **Keywords:** NGC 2014, NGC 2020, Wolf-Rayet cone, oxygen gas, 11000K, ejecta, radiation, UQFF **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

Variant 2 of the NGC 2014/2020 analysis focuses on the dominant Wolf-Rayet star cone structure in NGC 2020 and the oxygen-rich hot gas at $T \approx 11000$ K forming the characteristic blue nebulosity. The WR stellar winds at 3000 km/s create expanding ring nebulae visible in [O III] 5007 Å emission.

1. System Parameters

Parameter	Value	Description
M_{WR}	$2 \times 10^5 M_\odot$	WR-dominated region mass
r_{cone}	1.2×10^{16} m	WR cone radius
L_{WR}	$2 \times 10^5 L_\odot$	WR stellar luminosity
v_{eject}	3000 km/s	WR wind ejecta velocity
T_O	11000 K	Oxygen gas temperature

2. Master UQFF Gravity Equation

$$g_{WRcone}(t) = \frac{GM_{WR}(t)}{r^2}(1+H_0 t)(1+f_{TRZ}) + \frac{L_{WR}}{4\pi r^2 c}\frac{\rho_{WR}}{m_H} + q(v \times B)\left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-12}$$

2.1 WR Mass Loss

$$M_{WR}(t) = M_0(1 + 50\, e^{-t/\tau_{WR,SF}})$$

2.2 Radiation Pressure on Oxygen Cone

$$a_{rad,WR} = \frac{L_{WR}}{4\pi r_{cone}^2 c} \cdot \frac{\rho_{WR}}{m_H}$$

This drives the characteristic OIII ring nebula expansion.

References

- Crowther (2007), Wolf-Rayet stellar winds
- UQFF Framework, Star-Magic Session 175

Whitepaper auto-generated by _gen_whitepapers_702_715.py – Star-Magic Session 175